

Imipenem exposure influences the expression of quorum-sensing receptor *sdiA* in *Escherichia coli*

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Abstract

The increasing trend of carbapenem resistance amongst *Escherichia coli* poses a major public health crisis and requires active surveillance of resistance mechanisms to control the threat. Quorum-sensing system plays a role in bacterial resistance to antibiotics. Quorum sensing is a cell–cell communication system where bacteria alter their gene expression in response to specific stimuli. Here, in this study we investigated the transcriptional response of quorum-sensing receptor, *sdiA* in *E. coli* under sub-inhibitory concentration of carbapenem in the presence of quorum-sensing signal molecules. Two *E. coli* isolates harbouring *bla*_{NDM} were subjected to treatment with 10% Sodium Dodecyl Sulphate (SDS) for 20 consecutive days of which *bla*_{NDM} encoding plasmid was successfully eliminated from one isolate. Both the wild type and the cured mutant were then allowed to grow under eight different inducing conditions and the transcriptional response of *sdiA* gene was studied by quantitative real-time Polymerase Chain Reaction (PCR) method. We found different response levels of *sdiA* in wild type and cured mutant under exogenous AHL and imipenem and when co-cultured with *Pseudomonas aeruginosa* under imipenem stress. This study highlighted that sub-inhibitory concentration of imipenem in combination with AHL is acting as a signal to SdiA, a quorum-sensing receptor in *E. coli*.

Keywords: Imipenem; *sdiA* *Escherichia coli*; exposures; expression; AHL

Introduction

The emergence and spread of carbapenem resistance in gram-negative bacteria such as *Escherichia coli* and *Pseudomonas aeruginosa* has narrowed down treatment options as carbapenems are considered one of the most reliable drugs for treating infections caused by these bacteria (Nordmann and Poirel 2019). The increasing trend of carbapenem resistance amongst these bacteria poses a major public health crisis and requires active surveillance of primary and secondary resistance mechanisms to control the threat (Brink 2019). Quorum-sensing system plays a role in bacterial resistance to antibiotics. Quorum sensing is a cell–cell communication phenomenon where bacteria engage in chemical signaling and adjust gene expression accordingly (Turan et al. 2017). The quorum-sensing phenomenon involves a typical system composed of LuxI synthase and LuxR transcriptional regulator protein. The LuxI synthase synthesizes Acyl Homoserine Lactone (AHL) molecules which bind to their cognate LuxR receptor protein (Papenfert and Bassler 2016, Waters and Bassler 2005). *Pseudomonas aeruginosa*, a model quorum-sensing organism has two quorum-sensing systems viz. *las* and *rhl* system which code for synthase and receptor proteins; *lasI* and *rhlI* synthesize N-oxododecanoyl homoserine lactone (C12AHL) and N-butanoyl Homoserine Lactone (C4AHL) molecules, respectively (Paczkowski et al. 2017). Studies reveal that many gram-negative bacteria including *E. coli*, *Salmonella*, and *Klebsiella* do not harbour any LuxI homolog, rather these bacteria encode a LuxR homolog which re-

sponds to AHLs synthesized by other bacteria (Lindsay and Ahmer 2005, Ahmer 2004). The quorum-sensing system described in *E. coli* includes an SdiA regulator, a LuxR homolog, which senses AHLs synthesized by other bacteria and also AHLs in the surrounding environment (Smith et al. 2011). *Pseudomonas aeruginosa* and *E. coli* are known to be the major causative agents of nosocomial infections caused by gram-negative bacteria. These are also associated with biofilm forming medical devices and catheter-associated urinary tract infections (Culotti and Packman 2014). Many interesting lines of research are currently focused on knowing how antibiotics may act as stimuli to affect inter-species bacterial cell–cell communication systems and the physiological responses that are affected as a result (Skindersoe et al. 2008, Lu et al. 2017). Given the alarming rate of carbapenem resistance in *E. coli* and *P. aeruginosa*, it would be interesting to focus on knowing whether carbapenems at sub-lethal doses may act as signal molecules to regulate quorum sensing in these bacteria. *E. coli* and *P. aeruginosa* are often found to occupy the same niche and *E. coli* being a quorum-sensing orphan organism has the ability to eavesdrop the quorum-sensing signal molecules released by other bacteria in the surrounding environment (Miller et al. 2018). Since both AHL and antibiotics are secreted in the surrounding environment, it would be interesting to know their combined effect on quorum-sensing regulatory genes in bacteria. Therefore, here in this study, we investigate whether carbapenem in the presence of quorum-sensing signal molecules influences the transcriptional response

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Table 1. The list of acyl homoserine lactones.

SI No.	Compound
1.	N butanoyl homoserine lactone
2.	N Oxopentanoyl homoserine lactone
3.	N hexanoyl homoserine lactone
4.	N heptanoyl homoserine lactone
5.	Octanoyl homoserine lactone
6.	Nonanoyl homoserine lactone
7.	Decanoyl homoserine lactone
8.	Undecanoyl homoserine lactone
9.	Oxododecanoyl homoserine lactone
10.	(S)-3-Oxo-N-(2-oxotetrahydrofuran-3-yl)tridecanamide
11.	3-oxo-N-[(3S)-2-oxotetrahydrofuran-3-yl]tetradecanamide
12.	N-cis-Hexadec-9Z-enoyl-L-homoserine lactone
13.	N-[(3S)-Tetrahydro-2-oxo-3-furanyl]octadecanamide
14.	N-(3-hydroxy-eicosanoyl)-homoserine lactone

of quorum-sensing receptor, *sdiA* in *E. coli* which acts as a transcriptional regulator of many functional genes.

Materials and methods

The study was performed with seven carbapenem non-susceptible clinical isolates of *E. coli* which were obtained as secondary sources from Silchar Medical College and Hospital, Silchar, a tertiary referral care hospital. These isolates were selected on the basis of their MICs against imipenem and meropenem; resistance towards cephalosporins (ceftazidime, ceftriaxone, and cefepime) and aminoglycosides (gentamycin and tobramycin) and the presence of either of carbapenemase genes.

Molecular docking of *sdiA* with AHLs

The molecular docking of *sdiA* protein of *E. coli* was performed with different Quorum Sensing (QS) signal molecules viz. AHLs. The list of AHLs is given in Table 1. The receptor *sdiA* having PDB ID 4LFU was docked with ligands (AHLs) using software FlexX. The respective structures of the ligands were retrieved from PubChem database. The Structure Data File (SDF) file of all the ligands was loaded in FlexX software as docking library and was docked with the amino acids in the binding site of *SdiA* using default parameters. The output file gave the energy values in Kcal/mol and for each docked molecule, this value was noted down.

Plasmid elimination assay

The elimination of plasmids from the bacterial isolates harbouring *bla_{NDM}* was carried out using 10% of Sodium Dodecyl Sulphate (Paul et al. 2020). A volume of 10 µl of an overnight culture of bacterial cells was transferred to 5 ml fresh Luria Bertani Broths containing 10% (w/v) SDS and was incubated at 37°C in a shaker incubator at 220 rpm. Cells were plated on Luria Bertani (LB) agar supplemented with 1 µg/ml imipenem after each SDS treatment. Plasmid elimination was estimated by calculating the number of cells growing on LB with and without imipenem. Furthermore, PCR assay was carried out after each treatment to check for the loss of *bla_{NDM}* gene. MIC of imipenem for the cured mutant was checked by broth dilution method and the results were compared against their untreated wild type.

Table 2. Different inducing conditions under which the wild and the cured mutant were grown.

Bacterial Isolates	Growth conditions
<i>Escherichia coli</i> (wild type) and <i>E. coli</i> (mutant)	Normal growth conditions (laboratory condition)
	With 1 µg/ml of Oxododeconyl homoserine lactone (C12AHL)
	With 1 µg/ml of Butanoyl homoserine lactone (C4AHL)
	With both C4AHL and C12AHL at 1 µg/ml each, respectively
	With 1 µg/ml imipenem
	With both imipenem and C12AHL at 1 µg/ml each, respectively
	With both imipenem and C4AHL at 1 µg/ml each, respectively
	With imipenem, C12AHL, and C4AHL at 1 µg/ml each, respectively

Transcriptional response analysis of *sdiA* gene under different inducing conditions

The transcriptional response of *sdiA* gene in both the wild type and the cured mutant *E. coli* strains was studied under different inducing conditions by quantitative real-time PCR method. Both the wild type and the cured mutant were allowed to grow under eight different inducing conditions which are mentioned in Table 2.

Both the wild type and the cured mutant were allowed to grow up to the early log phase under the mentioned inducing conditions in fresh LB broths. The cells were grown at 37°C with shaking at 200 rpm until OD_{600nm} of 0.8–1.0. Cell pellets obtained after centrifugation were then used for extraction of mRNA. mRNA was extracted from the isolates using the RNeasy kit (Qiagen, Germany) following the manufacturer's protocol. The extracted mRNA was then immediately reverse-transcribed to cDNA using the Quanti Tect Reverse Transcription kit (Qiagen, Germany) following the manufacturer's protocol. The synthesized cDNA was then assessed by running a semi-quantitative PCR using oligonucleotide primers *sdiA* (F) 5'-AAGGACAAAAAGCCCAACGC-3', *sdiA* (R) 5'-AAGCACAGCCGTTATGGGAA-3' designed in the present study. The transcriptional response of *sdiA* gene was assessed in the wild type and cured mutant in Step One Plus real-time detection system (Applied Biosystem, USA) with the synthesized cDNA as templates. The target samples were the strains (wild type and cured mutant type) cultured under different inducing conditions while the reference samples were the strains (wild type and cured mutant) cultured under normal growth conditions. The experiment was performed in triplicate and the ct values were obtained. The relative quantity (RQ) of the expression of *sdiA* gene in the isolates was evaluated using the $\Delta\Delta$ ct method. 30S ribosomal protein subunit *rpsL* gene of *E. coli* was also used in parallel as an internal control throughout the reactions.

Transcriptional response analysis of *sdiA* gene in the wild type and cured mutant co-cultured with *P. aeruginosa*

A carbapenem-resistant clinical isolate of *P. aeruginosa* harbouring *bla_{NDM}* gene and the wild type and the cured *E. coli* mutant were

separately cultured overnight in fresh LB broths. The overnight cultures were diluted to a final optical density of 0.05 and inoculated at a 1:1 ratio in two fresh 5 ml LB broths of which one was supplemented with 1 µg/ml imipenem (Culotti and Packman 2014, Gao et al. 2021). Since *P. aeruginosa* endogenously synthesizes AHL, exogenous AHLs were not added to the growth media. The co-cultures were allowed to grow up to log phase and mRNA was extracted using the RNeasy kit (Qiagen, Germany) and the extracted mRNA was then immediately reverse-transcribed to cDNA using the Quanti Tect Reverse Transcription kit (Qiagen, Germany) following the manufacturer's protocol. The transcriptional response of *sdia* gene was assessed in the wild type and cured mutant in Step One Plus real-time detection system (Applied Biosystem, USA) with the synthesized cDNA as templates.

Statistical analysis

A two-way Analysis of Variance (ANOVA) was carried out in SPSS software to analyse the effects of different inducing conditions and strain type on the expression (RQ) of *sdia* gene. Differences were considered statistically significant at when P-value ≤ .05.

Results

The seven carbapenem non-susceptible clinical isolates of *E. coli* obtained for the present study had MIC values above breakpoint for imipenem, meropenem, and ertapenem. Two isolates harboured *bla_{NDM}* gene. The study isolates were also resistant to all the tested aminoglycosides (gentamycin and tobramycin) and cephalosporins (ceftazidime, ceftriaxone, and cefepime). No other classes of carbapenemase genes were found in any isolate.

Molecular docking of SdiA with AHLs

The docking program FlexX was used to dock different AHL molecules with the binding site of SdiA. A total of 14 AHL molecules were retrieved as SDF files from PubChem database and docked with the amino acids in the binding site of SdiA. All the 14 AHLs formed docking complex with SdiA and their binding energies were analysed by LeadIT. Among the docked AHLs, it was revealed that C4AHL had the best docking score. The docking scores of the docked complexes are given in Table 3. This study selected C4AHL and C12AHL for further *in vitro* studies as these two AHLs are synthesized and released by *P. aeruginosa* and the study investigated interaction between *E. coli* and *P. aeruginosa*.

Elimination of plasmid from the isolates harbouring *bla_{NDM}*

Two isolates harbouring *bla_{NDM}* were subjected to treatment with 10% SDS for 20 consecutive days. *bla_{NDM}* encoding plasmid was successfully eliminated from one isolate after the 18th treatment. PCR analysis of the cured mutant revealed the successful elimination of *bla_{NDM}* encoding plasmid from one isolate after 18th consecutive treatment thus confirming plasmid curing. The MIC of imipenem and meropenem for the cured mutant against its untreated wild type also reduced from 64 µg/ml to 4 µg/ml thereby indicating successful plasmid curing.

Up-regulation of *sdia* gene in wild type and cured mutant in response to C4AHL, C12AHL, and imipenem in combination

The transcriptional response of *sdia* gene in both the wild type and the cured mutant *E. coli* strains was studied under differ-

Table 3. The docking scores of the docked complexes.

SI No.	Ligands	Protein	Total score (Kcal/mol)
1.	N butanoyl homoserine lactone	<i>sdia</i> (PDB ID: 4LFU)	-14.0357
2.	N Oxopentanoyl homoserine lactone		-13.06
3.	N hexanoyl homoserine lactone		-11.9108
4.	N heptanoyl homoserine lactone		-11.98
5.	Octanoyl homoserine lactone		-9.892
6.	Nonanoyl homoserine lactone		-10.4779
7.	Decanoyl homoserine lactone		-9.5012
8.	Undecanoyl homoserine lactone		-10.33
9.	Oxododecanoyl homoserine lactone		-10.70
10.	(S)-3-Oxo-N-(2-oxotetrahydrofuran-3-yl)tridecanamide		-9.76
11.	3-oxo-N-[(3S)-2-oxotetrahydrofuran-3-yl]tetradecanamide		-14.0017
12.	N-cis-Hexadec-9Z-enoyl-L-homoserine lactone		-11.5524
13.	N-[(3S)-Tetrahydro-2-oxo-3-furanyl]octadecanamide		-8.654
14.	N-(3-hydroxy-eicosanoyl)-homoserine lactone		-6.3368

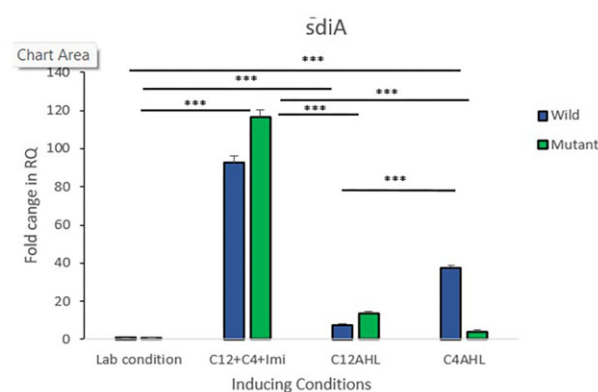


Figure 1. Transcriptional response of *sdia* gene in wild and cured mutant *E. coli* when grown under C12AHL and C4AHL and C12AHL, C4AHL, and imipenem in combination at 1 µg/ml, respectively. The fold change in relative quantity (RQ) values of the expression of *sdia* gene is plotted on the Y-axis. The error bars represent Standard Deviation (SD). Asterisks designate significance (*P < .05, **P < .01, and ***P < .001).

ent inducing conditions by quantitative real-time PCR method. It was observed that the response of *sdia* gene was up-regulated in both the wild type and cured mutant when these were grown with 1 µg/ml C4AHL, C12AHL, and imipenem in combination, respectively. The response of the gene increased multiple-folds in comparison to their response under normal growth conditions (Fig. 1).

sdia gene response is up-regulated in response to C4AHL in combination with imipenem and C12AHL in combination with imipenem

The transcriptional response of *sdia* gene was found to be up-regulated in both wild type and cured mutant when these were grown with exogenous C4AHL and imipenem, both at 1 µg/ml concentration, respectively. The response was also found to be enhanced in both the wild type and cured mutant when these were grown with exogenous C12AHL in combination with imipenem both at 1 µg/ml concentration, respectively (Fig. 2).

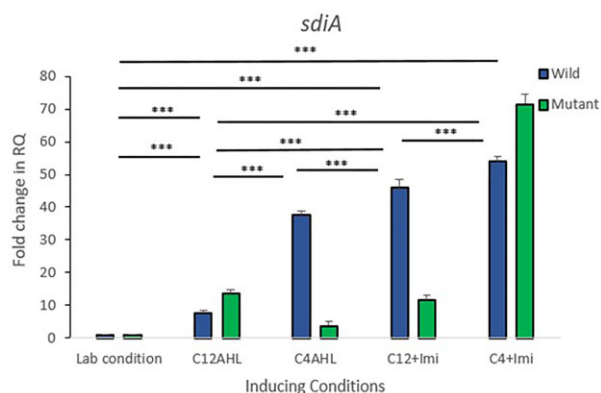


Figure 2. Transcriptional response of *sdiA* gene in wild and cured mutant *E. coli* when grown under C12AHL and C4 AHL; C4AHL in combination with imipenem and C12AHL in combination with imipenem at 1 µg/ml, respectively. The fold change in relative quantity (RQ) values of the expression of *sdiA* gene is plotted on the Y-axis. The error bars represent SD. Asterisks designate significance (* $P < .05$, ** $P < .01$, and *** $P < .001$).

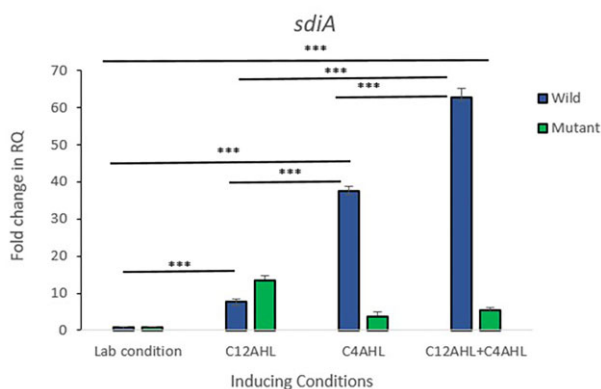


Figure 3. Transcriptional response of *sdiA* gene in wild and cured mutant *E. coli* when grown under C12AHL and C4AHL and C12AHL and C4AHL in combination. The fold change in relative quantity (RQ) values of the expression of *sdiA* gene is plotted on the Y-axis. The error bars represent SD. Asterisks designate significance (* $P < .05$, ** $P < .01$, and * $P < .001$).

Response of *sdiA* gene is up-regulated in response to C4AHL and C12AHL in combination

The transcriptional response of *sdiA* gene was found to be up-regulated in both wild type and cured mutant when these were grown with exogenous C4AHL and C12AHL in combination at 1 µg/ml concentration each. The response increased multiple-folds in comparison to their response under normal growth conditions. However, the enhanced response of *sdiA* gene was more pronounced in the wild type in comparison to the cured mutant (Fig. 3).

The response of *sdiA* gene did not vary among the wild and the mutant isolate under 1 µg/ml imipenem stress

We compared the transcriptional response of *sdiA* gene in wild type and cured mutant grown under 1 µg/ml imipenem stress to wild type and cured mutant grown under normal growth conditions and no significant difference (P -value = 1) was found in the response of *sdiA* gene among the wild type and cured mutant in comparison to their response under normal growth condition (Fig. 4).

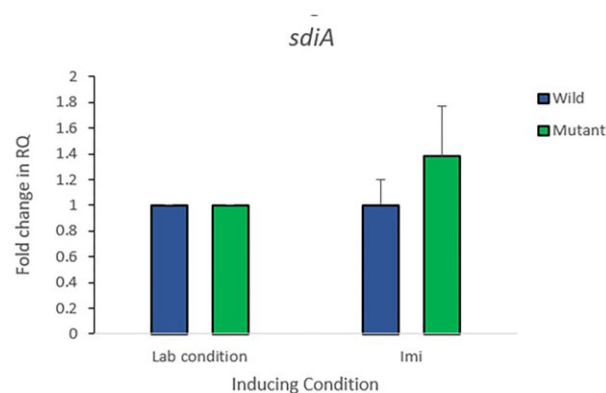


Figure 4. Transcriptional response of *sdiA* gene in wild and cured mutant *E. coli* when grown under 1 µg/ml imipenem. The fold change in relative quantity (RQ) values of the expression of *sdiA* gene is plotted in the Y-axis. The error bars represent SD.

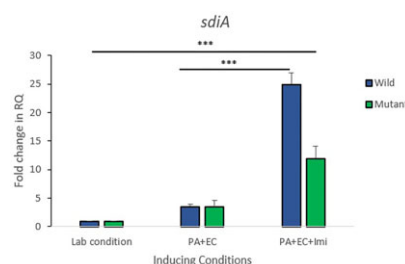


Figure 5. Transcriptional response of *sdiA* gene in wild and cured mutant *E. coli* when co-cultured with *P. aeruginosa*. The fold change in relative quantity (RQ) values of the expression of *sdiA* gene is plotted on the Y-axis. The error bars represent SD. Asterisks designate significance (* $P < .05$, ** $P < .01$, and * $P < .001$).

Up-regulation of *sdiA* gene in wild and cured mutant when co-cultured with *P. aeruginosa*

Wild type and cured mutant *E. coli* were respectively co-cultured with a carbapenem-resistant *P. aeruginosa* isolate in a 1:1 ratio in two separate 5 ml LB broths supplemented with 1 µg/ml imipenem. Since *P. aeruginosa* endogenously synthesizes AHL, no exogenous AHL was added to the growth media. The transcriptional response of *sdiA* gene was found to be up-regulated in comparison to the response in normal growth conditions. However, the response of *sdiA* gene did not vary among wild type and cured mutant. To determine if imipenem at 1 µg/ml modifies the transcriptional response of *sdiA* gene, we co-cultured the wild type and cured mutant, respectively with *P. aeruginosa* in LB broth supplemented with 1 µg/ml imipenem. It was observed that the transcriptional response of *sdiA* gene increased multiple-folds in both the wild type and the cured mutant (Fig. 5).

Statistical analysis

Two-way ANOVA analysis revealed that there was a statistically significant interaction between the effects of different inducing conditions and strain type ($P \leq .0001$) on RQ of *sdiA* gene (Fig. SF1). The whole summary is provided in Table ST1. Tukey's Honestly Significant Difference (HSD) post hoc test for pairwise comparisons (Inducing conditions and RQ) was also carried out and the statistically significant comparisons are shown on the graphs (Figs 1–5) as asterisk.

Discussion

The present study was performed to explore transcriptional response of *sdiA* gene in wild type and cured mutant *E. coli* grown under different inducing conditions. In this study we used wild type and cured mutant *E. coli* strains cultured under normal growth conditions as controls because these were grown without any exogenous AHL or antibiotic; besides, we wanted to know how the response of *sdiA* gene varied among wild and cured mutant *E. coli* when grown under different inducing conditions. In nature, most bacteria exist within mixed populations and interaction between population members includes responses to nutritional compounds, antibiotics, and chemical signals (Little et al. 2008, Phelan et al. 2012). AHL molecules synthesized by *P. aeruginosa* are used as signals by many gram-negative bacteria including *E. coli* and *Salmonella*. *sdiA*, a transcription factor of the LuxR family responds to AHLs produced by other bacteria and also synthetic AHLs (Smith et al. 2011, Mayer et al. 2023). Firstly, in this study, computational insights were performed by docking different AHLs in the active of *sdiA*. Among the docked AHLs it was revealed that C4AHL had the highest docking score. Subsequently, wild type and cured mutant were grown with exogenous C4AHL, C12AHL, and imipenem and different comparisons were made to obtain relative quantification of *sdiA* gene with respect to wild type and cured mutant cultured under normal growth conditions. We found different expression levels in wild type and cured mutant under exogenous AHL and imipenem as well as when co-cultured with *P. aeruginosa* under imipenem stress.

In the present study, we found the up-regulation of *sdiA* gene in wild type and cured mutant in response to exogenous AHLs—C4AHL and C12AHL both singly and in combination. In response to C4AHL and C12AHL added in combination, the expression level of *sdiA* gene in wild type and cured mutant increased further. Similar results were reported by studies where the response of *sdiA* gene was enhanced in response to C6AHL and indole (Lindsay and Ahmer 2005, Sabag-Daigle et al. 2012).

When the wild type and the cured mutant were allowed to grow with imipenem stress at 1 µg/ml, no significant change in expression of *sdiA* gene was observed with respect to their response in normal growth conditions (P -value = 1), which is in contrary to studies who have reported antibiotics enhancing the expression of luxR homologs (Skindersoe et al. 2008, Enoki et al. 2015). However, an interesting observation of this study is the up-regulation of *sdiA* gene in both wild type and cured mutant in response to imipenem in combination with C4AHL and C12AHL. Maximum expression was observed when the strains were grown with C4AHL, C12AHL, and imipenem, all in combination at 1 µg/ml, respectively. The expression increased 92-folds and 116-folds in wild type and cured mutant, respectively. This result suggests that imipenem in combination with AHLs influences the expression of *sdiA* gene. Similar results were obtained when the wild type and the cured mutant were co-cultured with *P. aeruginosa* under imipenem stress. There was no significant change in expression of *sdiA* gene when the strains were co-cultured with *P. aeruginosa* and allowed to grow under normal growth conditions (P -value = .89). However, the expression increased multiple-folds when grown under imipenem stress. A few previous studies have reported that antibiotics at sub-inhibitory concentration regulate the expression of quorum-sensing genes in response to antibiotics, AHL, and non-AHL signal molecules (Liu et al. 2013, Nguyen et al. 2015). Interestingly in this study, imipenem singly did not change the expression of *sdiA* gene but influenced its expression whenever added in combination with AHL.

In conclusion, the findings of the present study elucidated the enhanced response of *sdiA* gene in both wild type and cured mutant *E. coli* in response to the sub-inhibitory concentration of imipenem in combination with AHL (C4AHL and C12AHL), respectively. The study underscores that sub-inhibitory concentration of imipenem in combination with AHL is acting as signal to *SdiA*, a quorum-sensing receptor in *E. coli*. According to our knowledge, this is the first report highlighting the influence of imipenem in combination with AHL to enhance the expression of quorum-sensing receptor gene. Further investigations, both at the phenotypic and protein level, are required to unravel the precise mechanism by which imipenem along with AHL influence *sdiA*-regulated expression in the event of antibiotic resistance in *E. coli*.

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Author contributions

Chandrayee Deshamukhya (Investigation, Methodology, Writing – original draft), Ferdaus Begom (Investigation, Writing – review & editing), Bhaskar Jyoti Das (Formal analysis, Writing – review & editing), Monjur Ahmed Laskar (Resources – analysis tools), Sangeeta Goala (Resources – analysis tools), Manabendra Dutta Choudhury (Conceptualization, Resources), Debadatta Dhar (Chanda) (Conceptualization, Resources), and Amitabha Bhattacharjee (Conceptualization, Supervision)

Supplementary data

Supplementary data is available at *FEMSLE Journal* online.

Conflict of interest: None declared.

Data availability

The data underlying this article are available in the article.

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